FINAL SCENARIO PORTFOLIO

Capstone Report — AI-Assisted Scenario Design & Strategic Foresight

Pathways for the Digital Dollar

A Three-Scenario Portfolio for CBDC Deployment Stress-Testing, 2026–2030

Bledar Blake Zenuni

Strategic Futures Cell · Individual Capstone

AI-Assisted Scenario Design & Strategic Foresight

Dr. Frederic Lemieux, Ph.D.

Georgetown University — School of Continuing Studies

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Executive Summary

This portfolio assesses U.S. central bank digital currency deployment under conditions of irreducible uncertainty through 2030. It does not forecast a single future. It models three plausible pathways — each diagnosing a different load-bearing assumption of the institutional consensus path — and develops a scenario-aligned recommendation set.

Pathway Alpha (Managed Transition) is the baseline: phased CBDC rollout under sustained G10 coordination preserves commercial bank intermediation through 2030. Pathway Bravo (the 2028 Digital Bank Run) stresses the speed axis: a regional bank collapse activates digital-rail flight-speed dynamics that slow-flight policy guardrails were not calibrated to absorb. Pathway Charlie (Sovereign Fragmentation) stresses the coordination axis: gradual divergence of CBDC standards across blocs fractures cross-border settlement, with capital controls re-emerging as a second-order consequence.

Bravo's diagnostic value is the identification of speed of flight as Alpha's load-bearing constraint; Charlie's is standard-setting jurisdiction. Both are warnings, not forecasts.

Nine recommendations follow Module 6's typology — three Robust actions executable across all pathways, three Contingent actions paired with observable signposts, three Scenario-Specific actions calibrated to particular futures. The portfolio extends the Module 5 video deliverable (Pathway Bravo, classified retrospective) and is designed to be consulted operationally as well as read narratively.

PART I — THE DECISION CONTEXT

1. Strategic Decision Context

This report assesses U.S. central bank digital currency (CBDC) deployment under conditions of irreducible uncertainty. It does not forecast a single future. It models three plausible pathways through 2030 — Pathway Alpha (Managed Transition), Pathway Bravo (the 2028 Digital Bank Run), and Pathway Charlie (Sovereign Fragmentation) — and develops scenario-aligned

strategic recommendations for the institutions responsible for guiding the transition. The portfolio extends the methodology of the Module 5 deliverable, which presented Pathway Bravo as a Strategic Futures Cell retrospective assessment. Where the video carries one pathway, this report carries the set.

The Decision Problem

The relevant decision-makers are the Board of Governors of the Federal Reserve System, the U.S. Department of the Treasury, the Office of the Comptroller of the Currency, and the Federal Deposit Insurance Corporation, working in coordination with international counterparts through the Financial Stability Board and G10 central banking architecture. The strategic question is not whether CBDCs will exist — pilot programs are already underway across major economies. The question is under what governance architecture, at what adoption pace, and with what relationship to commercial-bank intermediation CBDCs are deployed. Each variation produces different systemic risks, different geopolitical exposures, and different demands on monetary policy transmission. The decision to be made is therefore not binary deployment, but the sequencing and guardrails of an already-occurring transition.

Key Challenges, Opportunities, and Risks

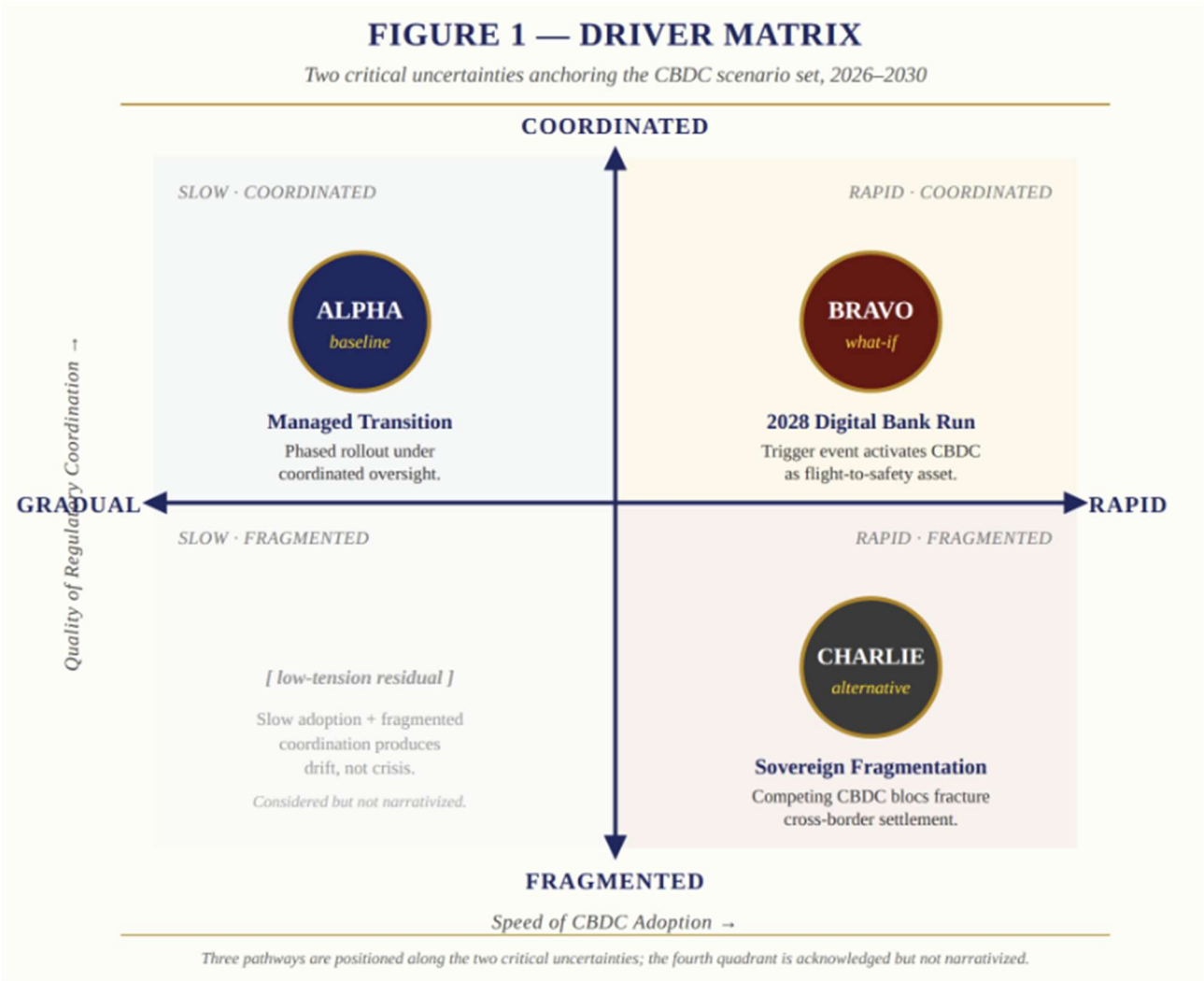
Three risk dimensions structure the scenario set. First, structural: digital payment rails enable instant flight-to-safety, breaking the slow-friction assumptions on which fractional-reserve banking depends — the Pathway Bravo failure mode. Second, geopolitical: CBDC technical standards function as soft power, and divergent standards across blocs risk fracturing the global payment system — the Pathway Charlie failure mode. Third, strategic opportunity: a well-sequenced rollout under coordinated oversight preserves intermediation while extending payments efficiency — the Pathway Alpha baseline. These pathways are not exhaustive; they are diagnostic of the principal axes of disagreement.

Scope

The portfolio addresses a four-year window from 2026 through 2030, with retrospective vantage from 2035 in the scenario narratives themselves. Geographic scope is U.S.-anchored and G10-coordinated, with global second-order effects modeled where they materially shape U.S. policy choices. Sectoral focus is on financial intermediation, monetary policy transmission, and payments infrastructure.

The Two Critical Uncertainties

Two critical uncertainties anchor the scenario set: the speed of CBDC adoption, and the quality of regulatory coordination across major economies. Figure 1 below positions the three pathways and identifies the residual fourth quadrant.



PART II — THE SCENARIO SET

2. The Three Pathways

The three pathways below share a single starting condition (the U.S. CBDC pilot operating at limited public adoption in 2026) and a single time horizon (through 2030). They diverge along the two critical uncertainties identified in Section 1: speed of CBDC adoption and quality of regulatory coordination. Pathway Alpha holds both axes favorable and serves as the baseline against which the others are read. Pathway Bravo holds coordination strong but stresses the speed axis through a 2028 trigger event. Pathway Charlie holds speed variable but stresses the coordination axis through gradual standards divergence. Read together, the set diagnoses the load-bearing assumptions of the baseline rather than predicting any single future.

TABLE 1 — COMPARATIVE SUMMARY OF THE THREE PATHWAYS

Three scenarios share a single starting condition (CBDC pilot, 2026) and time horizon (through 2030); they diverge along two critical uncertainties.

DIMENSION	PATHWAY ALPHA *Managed Transition · Baseline*	PATHWAY BRAVO *2028 Digital Bank Run · What-If*	PATHWAY CHARLIE *Sovereign Fragmentation · Alternative*
Trigger / inflection	Phased rollout; no triggering event	Regional bank collapse, Q1 2028 (SVB-pattern)	Bloc divergence on standards; gradual, no single trigger
Adoption pace	Gradual S-curve, 2027–2030; ~12–15% of M1 by 2030	Stable through 2027; sudden flight-to-safety in Q1–Q2 2028	Variable by bloc; aggregate adoption fragmented
Coordination quality	Strong; sustained through 2030	Strong pre-trigger; strained during cascade; restored by 2029	Erodes through 2027; structurally fragmented by 2029
Load-bearing failure mode	None activated	Speed of flight under digital rails	Standard-setting jurisdiction across blocs
Outcome by 2030	Hybrid steady-state; banks preserved; CBDC at retail scale	Consolidated banks; CBDC demoted to controlled backup	Three-bloc payment fracture; capital controls re-emerge

Pathway Alpha holds both axes favorable. Bravo stresses the speed axis. Charlie stresses the coordination axis. The set diagnoses Alpha's hidden dependencies rather than predicting any single future.

2.1 Pathway Alpha — Managed Transition (Baseline)

Pathway Alpha represents the rollout most aligned with current stated policy intentions across G10 central banks. It is designated the baseline because it contains no rupture event, not because it is the most likely outcome. Baseline in scenario planning means the trajectory absent shock — the pathway against which deviations are measured.

Drivers and dynamics: The Federal Reserve advances its CBDC pilot in measured phases between 2026 and 2028, expanding from invite-only access to opt-in public availability across major metropolitan areas. The load-bearing policy choice is announced in 2027: a per-individual

holdings cap, set at a level calibrated to permit retail payments efficiency without enabling material flight-to-safety dynamics. Coordinated G10 standards converge on interoperability protocols through 2028, ratified at the Financial Stability Board level. Public adoption follows a slow S-curve; CBDC reaches roughly twelve to fifteen percent of M1 by 2030, leaving commercial bank deposits as the dominant liquid store of value. Commercial banks adapt: deposit pricing adjusts modestly, intermediation margins compress, but the fractional-reserve model holds intact.

Narrative arc: The pathway's defining feature is what it does not contain. There is no triggering bank failure. There is no panic-speed deposit migration. There is no fragmentation of cross-border settlement architecture. The pilot expansion in 2026 is administratively unremarkable; the holdings cap announcement in 2027 generates policy-press coverage but no market dislocation; the international protocols ratified in 2028 are technical achievements rather than political events. By 2030, the system has matured into a stable hybrid: CBDCs handle a meaningful share of retail payment volume; commercial banks retain their core function as creators of credit through deposit-funded lending; central banks gain finer-grained instruments for monetary policy transmission.

Diagnostic value. Alpha is unspectacular by design. Its strategic utility lies precisely in what it withholds — and therefore in what the other two pathways reveal it has been quietly assuming. Bravo and Charlie are not alternative futures so much as they are stress tests of Alpha's hidden dependencies.

2.2 Pathway Bravo — The 2028 Digital Bank Run (What-If)

Pathway Bravo is the What-If pathway, explored in immersive narrative form through the Module 5 video [see Appendix B]. The video carries the story; this section carries the structural analysis.

Drivers and dynamics: Bravo shares its starting state with Alpha through late 2027 — same pilot, same caps, same coordination. The divergence comes from a single trigger: in early 2028, a regional bank collapse on the Silicon Valley Bank pattern materializes amid already-elevated rate-environment stress. The dynamic activated by the trigger is the one Alpha had been quietly assuming away — that depositors would not, in panic, treat the CBDC as a flight-to-safety alternative to commercial bank deposits. They do. The digital payment rails that gave the CBDC its retail efficiency advantage now give it a flight-speed advantage. Within weeks, deposit

outflows exceed forty percent across regional and mid-tier banks. The Alpha-era holdings caps were calibrated to slow-flight conditions; under panic-speed flight, they prove insufficient. CBDC wallets approach capacity limits, exposing infrastructure constraints that had not been visible at lower volumes.

Narrative arc, compressed: The cascade unfolds in identifiable phases through 2028. First quarter: the trigger event; equity markets fall; CBDC inflows surge. First through second quarter: contagion transmits through interbank markets; G10 central banks enter emergency coordination. Second quarter: banks close; lending halts; CBDC infrastructure strain becomes visible. Third quarter: a coordinated global policy response is implemented — holdings caps tightened, deposit guarantees expanded, emergency liquidity injections deployed. By 2029 and 2030, the system stabilizes, but it is permanently changed: the banking sector consolidates into fewer, stronger institutions, and the CBDC's role is repositioned from primary store of value to controlled backup. Figure 2 in Section 5 illustrates the modeled deposit trajectory.

Diagnostic value: Bravo's analytical contribution is the identification of speed of flight as Alpha's load-bearing assumption. The full narrative texture, broadcast register, and 2035 retrospective framing are carried by the Module 5 video, which functions as primary evidence for this pathway in the portfolio.

2.3 Pathway Charlie — Sovereign Fragmentation (Alternative)

Pathway Charlie tests Alpha's second hidden assumption: that international regulatory coordination will hold throughout the deployment window. Unlike Bravo, Charlie has no single trigger event. The failure mode is gradual divergence — the slow hardening of incompatible technical and governance standards across major economic blocs.

Drivers and dynamics: Three competing CBDC ecosystems emerge with structurally incompatible architectures. The dollar-anchored bloc (under the assumptions of this scenario, U.S.- and G7-aligned) develops a privacy-leaning, market-coordinated standard. The surveillance-architectural bloc (the People's Republic of China–led digital yuan ecosystem and its Belt-and-Road settlement partners, again under scenario assumptions) advances a state-coordinated standard with built-in transaction-level visibility. The regulation-led bloc (the European Union's digital euro project and aligned jurisdictions) advances a third pole, GDPR-aligned and rules-heavy. Cross-bloc settlement requires intermediary conversion layers that re-introduce the very friction CBDCs were intended to remove. Geopolitical alignment increasingly

determines payment infrastructure access; smaller economies are forced to choose blocs or attempt explicit neutrality, with patterns visible in the historical Swiss and Singaporean models.

Narrative arc: The divergence becomes visible in 2026 and 2027 as competing technical standards harden and interoperability working groups stall. In 2028, a first major cross-bloc settlement disruption arises from a sanctioned-entity test case that forces each bloc to formalize conversion controls. Through 2028 and 2029, capital controls quietly re-emerge across emerging markets caught between blocs. By 2030, the G10 coordination architecture is nominally intact but has become a forum for managing fragmentation rather than preventing it. Banking-sector outcomes diverge from Alpha and Bravo: institutions specialize by bloc; multi-bloc banks face compliance arbitrage costs that compress margins; smaller cross-bloc payment corridors proliferate but at significantly higher friction than the integrated payment architecture Alpha had projected.

Diagnostic value: Where Bravo exposes speed as a load-bearing constraint of Alpha, Charlie exposes standard-setting jurisdiction as the second. Both pathways are warnings about Alpha's hidden dependencies. Neither is a forecast; together, they bracket the conditions under which the baseline holds.

PART III — METHOD

3. Drivers, Assumptions, and Prompt Engineering

3.1 Cross-Scenario Drivers and Assumptions

The three pathways share a common foundation and diverge along specific driver assumptions, made explicit here:

- **Shared across all three pathways:** Underlying CBDC technology is assumed available and operationally deployable by 2026; a U.S. Federal Reserve CBDC pilot exists by 2026; no exogenous non-financial shock occurs within the time horizon; CBDCs are retail-accessible (not wholesale-only).

- **Pathway-specific assumptions:** Alpha assumes slow-flight depositor behavior, sustained G10 coordination, intermediation preserved through holdings-cap policy design. Bravo assumes panic-speed flight under digital rails, holdings caps calibrated to slow-flight conditions, and CBDC infrastructure capacity limits unforeseen at the policy stage. Charlie assumes geopolitical bloc divergence, incompatible technical standards, and capital control re-emergence as a second-order consequence.
- **Critical assumptions made explicit:** The portfolio does not model wholesale-only CBDC deployment, which would yield a fundamentally different scenario set. The portfolio assumes the U.S. dollar's reserve-currency status is contested but not displaced within the time horizon.

3.2 Prompt Engineering: Techniques Applied

The portfolio's AI-generated content (storyboard panels, classified frames, voiceover script, scenario scaffolding) was produced using five of the seven prompt design techniques in Bradshaw's typology (2025), with deliberate alignment to the analytical task taught in Module 2 (Lemieux, 2026a).

Role prompting anchored the scenario voice: Image-generation prompts framed the model as a producer of "stylized editorial illustration in the broadcast-news register"; the voiceover script was written for delivery in the role of "a senior analyst briefing the G10 Financial Stability Working Group in 2035." Role specification was load-bearing because it carried the epistemic register of the scenario into the medium. A different role prompt (e.g., "you are a news anchor reporting live") would have produced photoreal-leaning output that contradicts the scenario's counterfactual status.

Negative prompting enforced the anti-determinism guardrail. Every image-generation prompt included explicit prohibitions: "no photorealism, no real anchor faces, no real institutional logos, no recognizable city skylines, no real classification markings." This is the technique most directly aligned with Module 5's hyperrealism warning (slide 10). The negative prompt is what prevents the model from drifting into the very failure mode the methodology forbids.

One-shot / few-shot prompting enforced visual consistency across nine separate generations. A locked style string was constructed once and carried verbatim across every panel

generation, functioning as a shared exemplar. Without this technique, each panel would have read as a different artist's work; with it, the storyboard reads as a single piece.

Recursive prompting produced the portfolio's most consequential methodological catches. An early Panel 6 generation named the Financial Stability Board explicitly — a constraint violation. Recursive prompting (review, regenerate with archetypal substitutes) produced the corrected version. Panel 7 went through a similar recursive cycle when the first generation under-emphasized the closing thesis.

Chain-of-thought prompting structured the analytical scaffolding for the scenario design itself. The drivers-and-assumptions decomposition above, the load-bearing-failure-mode column in Table 1, and the cross-scenario diagnostic logic in Section 2 were all developed through step-by-step decomposition prompts rather than direct one-shot outputs.

Two further techniques were applied lightly. Retrieval-augmented generation: course slides for Modules 5 and 6 were uploaded to ground methodological alignment, and the Module 5 storyboard was uploaded for portfolio cross-checking. Meta-prompting was used in constructing the locked style string itself — a prompt designed to standardize subsequent prompts.

3.3 Full Prompts

The full text of all production prompts — locked style string, panel-by-panel generation prompts, classified cover and closing frame prompts, and voiceover script — is provided verbatim in Appendix A, with brief annotation indicating the technique each prompt embodies.

3.4 Improvements via Recursive Engineering

Two production iterations bear specific note. First, the Panel 6 regeneration: the first output named "Financial Stability Board" in both header and central composition; recursive review identified the constraint violation; the regenerated panel uses archetypal substitutes ("Coordinated Policy Response," "Global Stability Coordination") that preserve scenario integrity. Second, the voiceover script was abridged from approximately 850 words to approximately 220 words across two iterations to fit runtime under tooling constraints. Both are recursive-prompting outcomes that demonstrate prompt engineering as iterative discipline rather than single-shot specification.

PART IV — CRITICAL EVALUATION

4. Quality and Strategic Fit Evaluation

4.1 Module 4 Quality Criteria Applied

Each pathway is evaluated below against the Module 4 criteria: internal logic, bias awareness, novelty, and strategic utility. The format combines compact assessments with the analytical reasoning behind each rating.

Pathway Alpha — Managed Transition. Alpha's analytical contribution is reference-class, not insight. It exists to be the surface against which the other pathways' deviations register.

- **Internal logic — strong.** No rupture event; causal chain is short and well-supported by current G10 policy intentions.
- **Novelty — intentionally low.** Alpha represents the institutional consensus path.
- **Utility — high (as baseline).** Alpha makes Bravo's and Charlie's deviations legible.
- **Bias risk — status-quo bias** is built in by definition. The pathway assumes guardrails work as designed, which is precisely the assumption the other two pathways exist to challenge.

The bias is mitigated by Alpha's explicit framing as baseline-not-prediction in Sections 1 and 2. Without that framing, status-quo bias would compound into implicit forecasting.

Pathway Bravo — The 2028 Digital Bank Run. Bravo is the portfolio's strongest stress test. Its diagnostic value lies in identifying speed-of-flight as Alpha's load-bearing assumption.

- **Internal logic — strong.** The cascade follows documented patterns from the 2008 and 2023 banking stress events, extended with the digital-rail flight-speed driver inferred from observable CBDC pilot transaction data.
- **Novelty — high.** The argument that holdings caps calibrated to slow-flight become insufficient under digital-rail flight-speed is not yet mainstream policy thinking, despite being a structurally clean inference.

- **Utility — high.** Bravo is the pathway most directly useful for stress-testing the deposit-flight assumption embedded in Alpha.
- **Bias risk — availability bias.** The Silicon Valley Bank pattern is recent and salient, and its prominence in the trigger event may reflect what is vivid rather than what is most probable.

The Module 5 video [Appendix B] provides the immersive evidentiary form for this pathway and was produced under the anti-determinism guardrails documented in Section 3.

Pathway Charlie — Sovereign Fragmentation. Charlie tests Alpha's second hidden assumption: that international coordination holds. Its causal chain is longer and depends on more variables than the other two pathways.

- **Internal logic — moderate.** The pathway spans more variables than Alpha or Bravo, and assumes geopolitical bloc divergence that could resolve more favorably than modeled.
- **Novelty — high.** Framing CBDC standards as soft-power instruments and tracing the consequent payment-system fracture is underexplored in U.S. domestic policy discussion.
- **Utility — high.** Charlie is the pathway most useful for thinking past financial-domain failures into governance failures.
- **Bias risk — framing simplification.** The three-bloc structure flattens a more granular reality in which neutral and partially-aligned jurisdictions complicate cleaner classifications.

This bias is acknowledged in Section 2 through explicit hedging language. The "moderate" internal logic rating is intentional — claiming uniform "strong" ratings across all three pathways would itself be a bias signal.

4.2 Improvements Made Since Module 4

The portfolio extends Module 4 scenario-design work in three specific ways:

- **A third pathway (Charlie) was developed** to test the coordination-axis failure mode. Earlier scenario sets did not separate the speed-axis failure mode from the coordination-axis failure mode.

- **A cross-scenario diagnostic spine was added** — the load-bearing failure mode column in Table 1 — to make the scenario set's analytical contribution explicit rather than implied.
- **All driver assumptions were surfaced explicitly in Section 3.1,** rather than embedded inside narrative. This exposes the pathways to honest critique and prevents hidden dependencies from doing unacknowledged work.

4.3 Visual Strategy — Enhancements and Hindrances

The portfolio's visuals were chosen to do work the prose cannot do as efficiently.

Enhancements:

- **Driver matrix (Figure 1)** communicates the analytical structure of the scenario set in a single image; the position of each pathway against the two critical uncertainties is immediately legible.
- **Comparison table (Table 1)** lets the reader see the three scenarios as a set, which is the methodological move the portfolio is built around.
- **The Module 5 video** carries Bravo's narrative immersion at a sensory level prose cannot reach — broadcast register, palette progression, and 2035 retrospective framing operate beyond what text can do.
- **Locked editorial-illustration style** applied across nine generations enforced visual consistency that signals authored artifact rather than assembled fragments.

Hindrances and tradeoffs (named, not concealed):

- **Stylized illustration carries implicit confidence** even when explicitly counterfactual. The persistent "// SCENARIO NODE // CONFIDENTIAL //" overlay mitigates but does not eliminate this.
- **The 2×2 matrix simplifies a multi-dimensional uncertainty space.** The empty fourth quadrant partially acknowledges this but cannot represent the full degrees of freedom.
- **Table 1's compactness loses nuance** that the prose recovers; readers consulting only the table will miss the analytical work behind each cell.

None of these tradeoffs disqualify the visuals. Each is a real tradeoff against alternative formats with different costs, named here rather than concealed.

PART V — STRATEGIC RECOMMENDATIONS

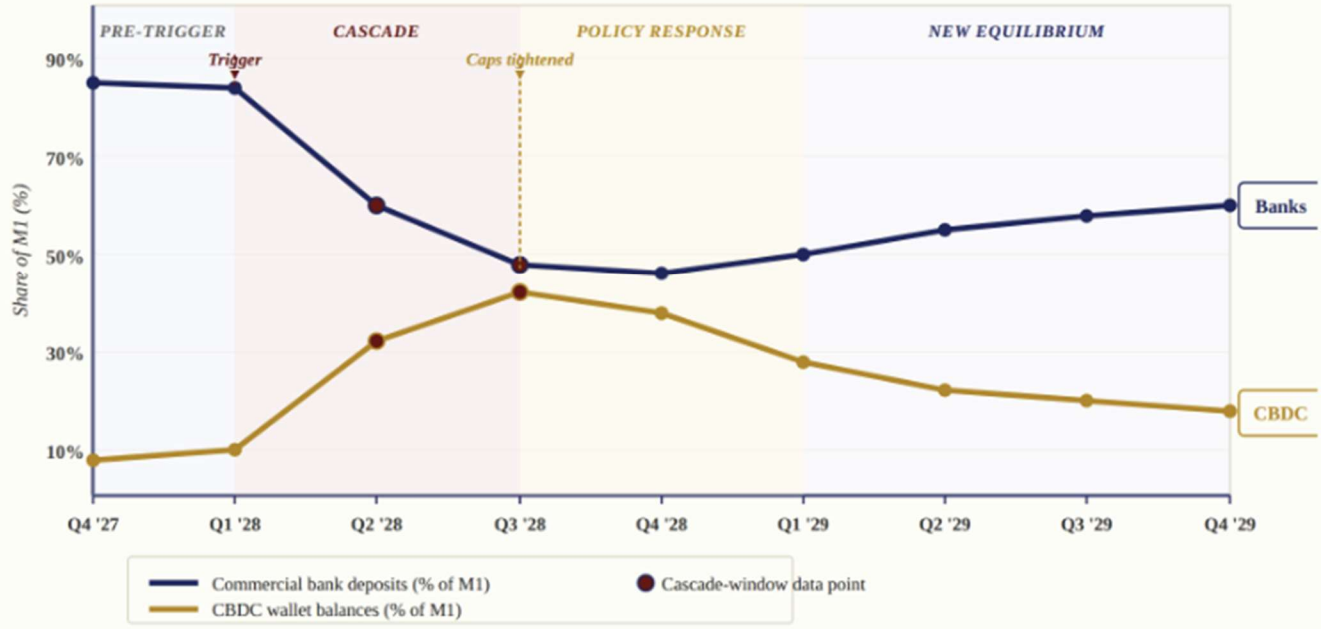
5. Scenario-Aligned Recommendations

Module 6 frames the relationship between scenarios and action directly: scenarios are maps; recommendations are the compass (Lemieux, 2026d, slide 16). This section translates the three pathways into a recommendation set organized around Module 6's three-part typology: Robust actions that pay off across all pathways, Contingent actions triggered by observed signposts, and Scenario-Specific actions that only make sense inside particular futures (Lemieux, 2026d, slides 5–6). Together they constitute the strategic counterpart to the scenarios in Section 2.

FIGURE 2 — PATHWAY BRAVO: MODELED DEPOSIT MIGRATION

Q4 2027 — Q4 2029 · % of M1 Money Supply · Illustrative scenario assumptions, not forecast

⚠ ILLUSTRATIVE — NOT FORECAST. Values reflect Pathway Bravo's modeled cascade under the assumptions of this scenario. They do not represent observed data or projected reality.



Two phase markers anchor the cascade: a Q1 2028 trigger event (regional bank collapse) and a Q3 2028 coordinated policy response (CBDC holdings caps tightened, deposit guarantees expanded). Pre-trigger and stabilization phases are shaded in cool tones; cascade and response phases in warmer tones. Aggregate share does not sum to 100%; remainder reflects other M1 components (currency in circulation, traveler's checks).

Figure 2. Pathway Bravo: modeled deposit migration, Q4 2027 — Q4 2029. Illustrative scenario assumptions, not forecast.

5.1 Robust Actions

Three actions are recommended regardless of which pathway materializes. Each preserves optionality, costs little relative to its hedging value, and pays off across the full scenario set.

- **Build CBDC stress-testing infrastructure inside the Federal Reserve's existing supervisory framework.** The capacity to model deposit migration under varying speed and coordination conditions is value-additive in Alpha (informs caps calibration), in Bravo (enables faster recognition of cascade onset), and in Charlie (supports cross-bloc settlement modeling). The infrastructure is reusable; the optionality is the asset.
- **Develop CBDC interoperability standards proactively, before Charlie-style fragmentation pressures harden.** U.S.-led standards-setting is a low-cost hedge that preserves leadership in Alpha, mitigates pressure in Charlie, and is irrelevant-but-harmless in Bravo. Standards work is path-dependent; the time to do it is before the divergent path locks in.
- **Maintain clear public communication that the CBDC is a complement to commercial bank deposits, not a replacement.** This anchors expectations under Alpha, slows panic-flight if Bravo's trigger materializes, and signals doctrine clarity to other blocs under Charlie. The cost is a communication discipline; the benefit is the quiet anchoring of depositor mental models.

5.2 Contingent Actions

Each contingent action triggers only when a specific observable indicator crosses a defined threshold. Indicators are summarized in Table 2 below.

- **If quarterly deposit-flight velocity exceeds the historical 75th percentile,** tighten CBDC holdings caps within 30 days. Direct mitigation for Bravo's trigger condition; the speed of the policy response must match the speed of the digital-rail flight that the trigger activates.
- **If two or more G10 central banks publicly diverge on CBDC technical standards,** activate a diplomatic interoperability working group with binding-track structure. Direct mitigation for Charlie's coordination erosion; the diplomatic cost rises sharply once

divergence is publicly committed, so the trigger should be early-signal rather than confirmation-stage.

- **If commercial bank deposit growth falls below the trailing twelve-month rate for three consecutive quarters,** expand emergency liquidity facility eligibility. A slower-burn precursor signal for Bravo; preempts the cascade rather than reacting to it.

5.3 Scenario-Specific Actions

These actions are calibrated to specific pathways and would be inappropriate or wasted under others.

- **Pathway Bravo: restructure FDIC deposit insurance to explicitly cover CBDC-driven flight scenarios.** Under current architecture, FDIC coverage assumes slow-flight conditions; under Bravo's panic-speed digital-rail flight, the insurance structure may not transmit confidence quickly enough to prevent contagion. Restructuring is invisible under Alpha (no flight to insure against) and inapplicable under Charlie (different failure mode entirely).
- **Pathway Charlie: establish bilateral CBDC conversion protocols with key non-aligned jurisdictions before bloc divergence hardens.** Switzerland and Singapore historically maintained currency neutrality through bilateral arrangements with rival blocs; the analogous CBDC architecture is achievable now and politically harder to negotiate later. Only meaningful under Charlie.
- **Pathway Alpha (sustained): advance CBDC adoption toward 20% of M1 by 2032 to capture retail-payment efficiency gains.** Under Alpha's stable conditions, the cap-and-pace policy choices that prevented disruption can be relaxed to capture the upside that motivated the project initially. Only sensible if Alpha holds; a category error if Bravo or Charlie materializes.

5.4 Recommendation Matrix and Signposts

Figure 3 organizes the nine recommendations against the three pathways in Module 6's recommendation typology format. Table 2 documents the observable indicators that should trigger each contingent action, following the signposts-and-triggers framework (Lemieux, 2026d, slide 8). Both visuals are designed to be consulted independently of the prose for operational use.

FIGURE 3 — RECOMMENDATION MATRIX

Nine recommendations organized by Module 6 typology (Robust · Contingent · Scenario-Specific) and by pathway

TYPOLOGY	PATHWAY ALPHA *Managed Transition*	PATHWAY BRAVO *2028 Digital Bank Run*	PATHWAY CHARLIE *Sovereign Fragmentation*
ROBUST *across all three*	**1.** Build CBDC stress-testing infrastructure inside the Federal Reserve's supervisory framework. **2.** Develop CBDC interoperability standards proactively, before fragmentation pressures harden. **3.** Maintain clear public communication: CBDC is a complement to commercial bank deposits, not a replacement.		
CONTINGENT *trigger-based*	— *No specific contingent triggers under stable baseline.*	**4.** If deposit-flight velocity exceeds historical 75th percentile → tighten CBDC holdings caps within 30 days. **5.** If commercial bank deposit growth falls below 12-month rate for 3 quarters → expand emergency liquidity facility eligibility.	**6.** If two or more G10 central banks publicly diverge on CBDC technical standards → activate diplomatic interoperability working group.
SCENARIO-SPECIFIC *pathway-bound*	**7.** Advance CBDC adoption to 20%+ of M1 by 2032 to capture retail-payment efficiency gains. *Only sensible if Alpha holds.*	**8.** Restructure FDIC deposit insurance to explicitly cover CBDC-driven flight scenarios. *Calibrated to panic-speed flight.*	**9.** Establish bilateral CBDC conversion protocols with key non-aligned jurisdictions before bloc divergence hardens. *Switzerland / Singapore patterns.*

Robust actions execute regardless of pathway. Contingent actions activate only when observable signposts (see Table 2) cross defined thresholds. Scenario-Specific actions are calibrated to particular pathways and would be category errors if applied under the wrong scenario.

Figure 3. Recommendation matrix: nine actions organized by Module 6 typology and pathway.

TABLE 2 — SIGNPOSTS AND TRIGGERS FOR CONTINGENT ACTIONS

Observable indicators that should trigger pre-defined responses, per Module 6 (Lemieux, 2026b, slide 8) framework

SCENARIO & ACTION	OBSERVABLE INDICATOR	RESPONSE TRIGGER
BRAVO Tighten CBDC holdings caps	*DEPOSIT-FLIGHT VELOCITY* Quarterly net outflow rate from commercial bank deposits to CBDC wallets, measured against a rolling 5-year baseline distribution.	Quarterly velocity **exceeds historical 75th percentile** for two consecutive measurement periods. Response: tighten holdings caps within 30 days.
BRAVO Expand emergency liquidity eligibility	*DEPOSIT GROWTH RATE* Year-over-year growth in commercial bank deposits, measured against trailing 12-month moving average.	Growth rate **falls below trailing 12-month rate for three consecutive quarters**. Response: expand emergency liquidity facility eligibility to mid-tier institutions.
CHARLIE Activate diplomatic working group	*STANDARDS DIVERGENCE* Number of G10 central banks publicly committing to incompatible CBDC technical or governance standards.	**Two or more G10 central banks publicly diverge** on CBDC technical standards. Response: activate diplomatic interoperability working group with binding-track structure.

Each contingent action is paired with an observable, measurable indicator and a defined trigger threshold. The discipline is to commit the response in advance, before pressure to delay accumulates.

Table 2. Signposts and triggers for contingent actions.

5.5 Closing

The three pathways together with the nine recommendations form the operational portfolio. The appendices that follow provide full prompt documentation, the Pathway Bravo video link, and references.

APPENDICES

Appendix A — Full AI Prompts

Prompts presented verbatim with annotation of the technique each embodies. Reference: Bradshaw (2025); Lemieux (2026a).

A.1 Locked Style String (One-Shot Exemplar)

Carried verbatim across all nine image generations in the Module 5 portfolio. Functioned as the one-shot exemplar enforcing visual consistency across separate generations.

> *Editorial illustration, muted broadcast-news aesthetic, slightly desaturated palette, isometric infographic elements where appropriate, NO photorealism, NO real logos, NO real anchor faces, NO named real institutions, NO real city skylines. Stylized and clearly illustrated — never confusable with real news footage. 16:9 widescreen aspect ratio. Cinematic but restrained — strategic-foresight register, not action-trailer. Persistent visual texture across all images so they feel like one continuous piece.*

Techniques embodied: One-shot prompting (locked exemplar), Negative prompting (NO clauses), Role prompting (strategic-foresight register).

A.2 Cover Frame Prompt (Classified Document)

> *Government/intelligence document aesthetic. Deep navy background. Times New Roman serif typography in white/off-white. Classification bars top and bottom (thin, dark gold accent). Composition: centered, formal, restrained. NO illustration of people or scenes — text-forward document cover. Top bar: // CONFIDENTIAL // SCENARIO ANALYSIS //. Main header: STRATEGIC FUTURES CELL / Retrospective Assessment — 2035. Subheader: SUBJECT: THE 2028 DIGITAL BANK RUN / A Seven-Year Retrospective. Distribution: NOT FOR ATTRIBUTION / METHOD EXPOSES SOURCES / G10 FINANCIAL STABILITY WORKING GROUP. Anti-determinism caveat: "The following is one of three scenarios developed in 2026 to stress-test CBDC deployment pathways. It was not a forecast. The events depicted did not occur."*

Techniques embodied: Role prompting, Negative prompting (no real classifications, no real seals).

A.3 Panel Generation Prompts (Panel 1 example)

Each of the seven story panels was generated using the locked style string above plus a panel-specific prompt. Panel 1 example below; Panels 2–7 followed the same structural template:

> *Panel 1 — Setting: Late 2027, Markets Calm. Same locked style. Stylized broadcast-news studio. Generic anchor figure (no recognizable face). Behind: large screen showing upward-trending stock chart in muted green, abstract data tickers. Soft warm studio lighting — confident, composed, pre-crisis. Color palette: warm muted blues and grays,*

slight amber highlights. Slightly oversaturated to evoke a "before times" archive feel. Lower-third chyron bar visible at bottom.

Techniques embodied: One-shot, Role prompting, Chain-of-thought (panel-by-panel narrative scaffolding).

A.4 Recursive Prompting — The Panel 6 Catch

The first generation of Panel 6 named "Financial Stability Board" explicitly — a constraint violation. The recursive correction prompt:

> *The previous Panel 6 was excellent in composition, palette, and tone — but it violated our locked style by naming a real institution ("Financial Stability Board") in the header AND on the central table. This breaks our anti-determinism rule. Regenerate keeping everything else the same: (1) Replace "Financial Stability Board" header with "COORDINATED POLICY RESPONSE." (2) Replace any text on the central table with "GLOBAL STABILITY COORDINATION." (3) Reduce header text weight so the three policy icons are primary. (4) Keep everything else from the previous version.*

Technique embodied: Recursive prompting (the iteration discipline that produced the corrected output).

A.5 Voiceover Script (Final Recorded Version)

Abridged from approximately 850 words to approximately 220 words across two iterations to fit ElevenLabs runtime and character constraints.

> *Late 2027. Markets were modeled as stable. The Federal Reserve's digital currency pilot had achieved limited adoption. Under the exercise's starting conditions… systemic risk appeared contained.*
>
> *Early 2028. The trigger: a regional bank collapse. Markets fell. Panic spread. But this time, depositors had a new option. With one tap, funds moved out of commercial banks… and into central bank digital currency wallets.*
>
> *Within weeks, deposit outflows exceeded forty percent. Liquidity pressures mounted. What began as a localized crisis… spread across the global financial system. Central bank leaders entered emergency coordination. Volatility reached levels last observed in 2008.*

Banks closed. Lending halted. The traditional model broke down. CBDC wallets reached capacity. The safe alternative… became part of the strain. Authorities responded. A coordinated global policy response: holding caps, expanded guarantees, liquidity injections.

By 2030, the system stabilized — but it was permanently changed. Fewer, stronger institutions. CBDCs remained… but their role had shifted. A controlled backup. The crisis exposed weaknesses. Yet… in this pathway… the system emerged with greater resilience. This was Pathway Bravo. One of three modeled. Its value was not predictive. Its value… was diagnostic.

Techniques embodied: Role prompting (2035 senior analyst register), Recursive prompting (two-pass abridgment).

A.6 Note on Module 3 Source Scenario Prompts

The original Module 3 What-If scenario construction (Systemic Disintermediation Crisis) was developed with prompt engineering scaffolding that has not been preserved verbatim. Rather than reconstruct prompts that were not preserved, this appendix limits itself to prompts cited with full fidelity. The Module 3 source scenario itself remains the analytical foundation referenced throughout this portfolio.

Appendix B — Pathway Bravo Video

The Module 5 video deliverable functions as the immersive narrative artifact for Pathway Bravo and is referenced throughout this portfolio as primary evidentiary form for that pathway.

- **Title:** *Strategic Futures Cell — Retrospective Assessment 2035: The 2028 Digital Bank Run*
- **Format:** MP4, 1920×1080, 1 minute 44 seconds
- **Production:** Image generation via OpenAI image model with locked style specification; voiceover via ElevenLabs ("Bill" voice, 2035 analyst register); assembly via Microsoft PowerPoint.
- **File:** *bravo_video_final.mp4 (submitted alongside this report)*

Appendix C — References

Course Materials

Lemieux, F. (2026a). Module 2: Prompt engineering for strategic foresight [Lecture]. AI-Assisted Scenario Design & Strategic Foresight, Georgetown University, School of Continuing Studies.

Lemieux, F. (2026b). Module 4: Scenario quality and evaluation criteria [Lecture]. AI-Assisted Scenario Design & Strategic Foresight, Georgetown University, School of Continuing Studies.

Lemieux, F. (2026c). Module 5: Visualizing and communicating AI-scenarios [Lecture]. AI-Assisted Scenario Design & Strategic Foresight, Georgetown University, School of Continuing Studies.

Lemieux, F. (2026d). Module 6: Strategic recommendations in scenario planning [Lecture]. AI-Assisted Scenario Design & Strategic Foresight, Georgetown University, School of Continuing Studies.

Source Scenario

Blake, B., & Gonzalez, K. (2026). Module 3 — CBDC What-If scenario: Systemic disintermediation crisis [Course assignment]. AI-Assisted Scenario Design & Strategic Foresight, Georgetown University, School of Continuing Studies.

Methodology

Bradshaw, P. (2025, February 19). 7 prompt design techniques for generative AI every journalist should know. Online Journalism Blog. https://onlinejournalismblog.com/2025/02/19/7-prompt-design-techniques-for-generative-ai-every-journalist-should-know/

AI Tools Used in Production

Anthropic. (2026). Claude (Opus 4.7) [Large language model]. https://www.anthropic.com/claude — Used for scenario design, methodological scaffolding, prompt engineering, drafting, and reflection.

ElevenLabs. (2026). Text-to-Speech — "Bill" voice profile [AI voice synthesis platform]. https://elevenlabs.io — Used for video narration.

Microsoft. (2026). PowerPoint (Microsoft 365) [Presentation and video assembly software]. Used for storyboard formatting and video assembly with MP4 export.

OpenAI. (2026). ChatGPT [Generative image model]. https://chat.openai.com — Used for storyboard panel visuals and classified frame designs.

Domain Sources

International Monetary Fund. (2024). Central bank digital currencies and financial stability [Fintech Note]. International Monetary Fund.

Normandin, M., Phaneuf, L., & Tremblay, P. (2025). Banking, central bank digital currencies, and macroeconomic stability [Working paper]. Cited in Module 3 source materials.

Rizwan, M. S., Ahmad, G., & Ashraf, D. (2020). Systemic risk: The impact of COVID-19. Finance Research Letters, 36, 101682.

U.S. House Republican Policy Committee. (2024). Brief on central bank digital currencies and financial sovereignty. Cited in Module 3 source materials.